TCSS 478/588

Bioinformatics Final Project Background & Motivation

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Winter 2026

MorPhiC (**Mo**lecula**r** **Ph**enotypes of Null Alleles **i**n **C**ells) https://morphic.bio/

Long term goals: To create a catalog of gene functions in human.

- Make null alleles (=~ “gene knockouts (KO)”)
- Of every gene (but start with ~1000 protein coding genes as a pilot)
- Optimize and scale up null generation strategies.
- Develop the analysis and visualization tools to facilitate the utilization of the MorPhiC data.

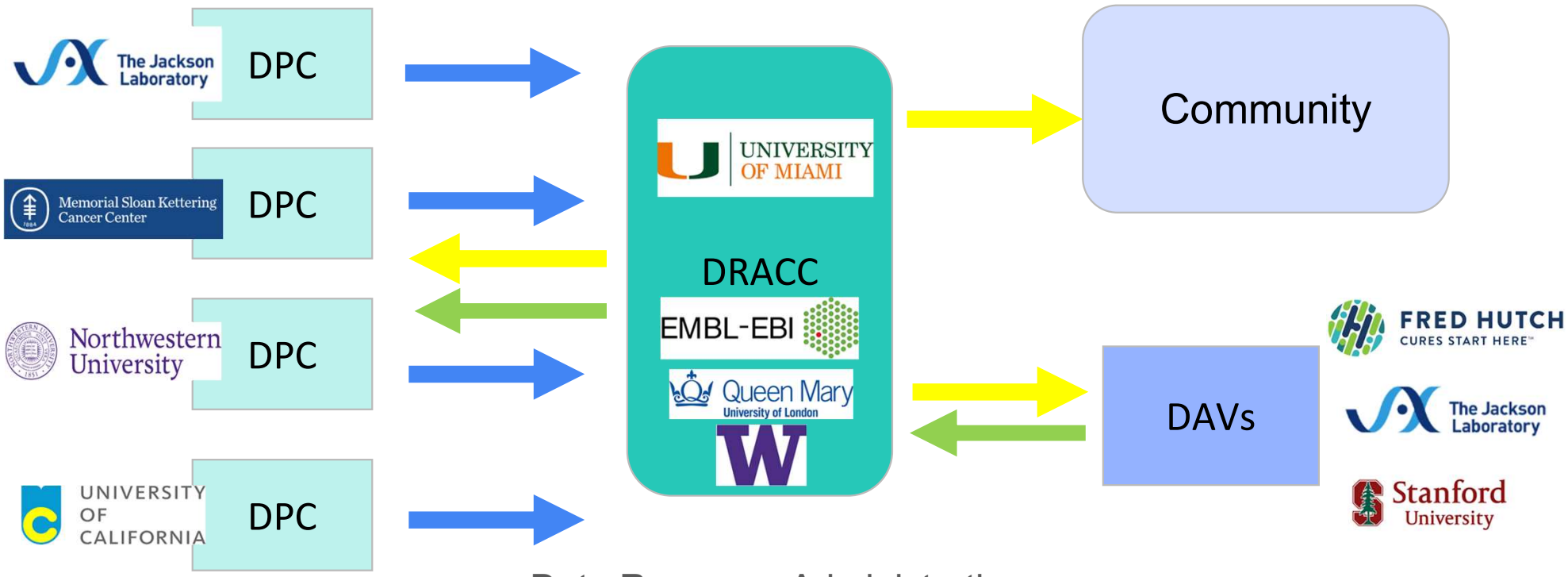
Marker paper "MorPhiC consortium: Towards functional characterization of all human genes” *Nature* (2025)

NIH press release (September 2022) https://www.nih.gov/news-events/news-releases/nih-initiative-systematically-investigate-establish-function-every-human-gene

NIH National Human Genome Research Institute

U24HG012674

MorPhiC Data Flow



Data Production Centers (DPCs)

Data Resource Administrative Coordination Center (DRACC)

U24HG012674

Data Analysis and Validation Centers (DAVs)

What does the UWT team do in MorPhiC?

Ling-Hong Hung

- Develop data processing analytical pipelines (https://github.com/morphic-bio)
  - Generalized open-source workflow that assigns feature barcodes, aligns and assigns cell barcodes for scRNA-seq/perturb-seq data
  - Faster, scalable end-to-end alternative to Cell Ranger
- Play an active role in data releases (https://morphic.bio/data/)
- Facilitating adoption
  - Biodepot training website at https://biodepot.github.io/training/
- Scalability of data processing
  - Schedulers with asynchronous execution (temporal.io)
  - Hybrid architecture with NSF super computers

Perspective

MorPhiC Consortium: towards functional characterization of all human genes

https://doi.org/10.1038/s41586-024-08243-w

Received: 31 October 2023

Accepted: 17 October 2024

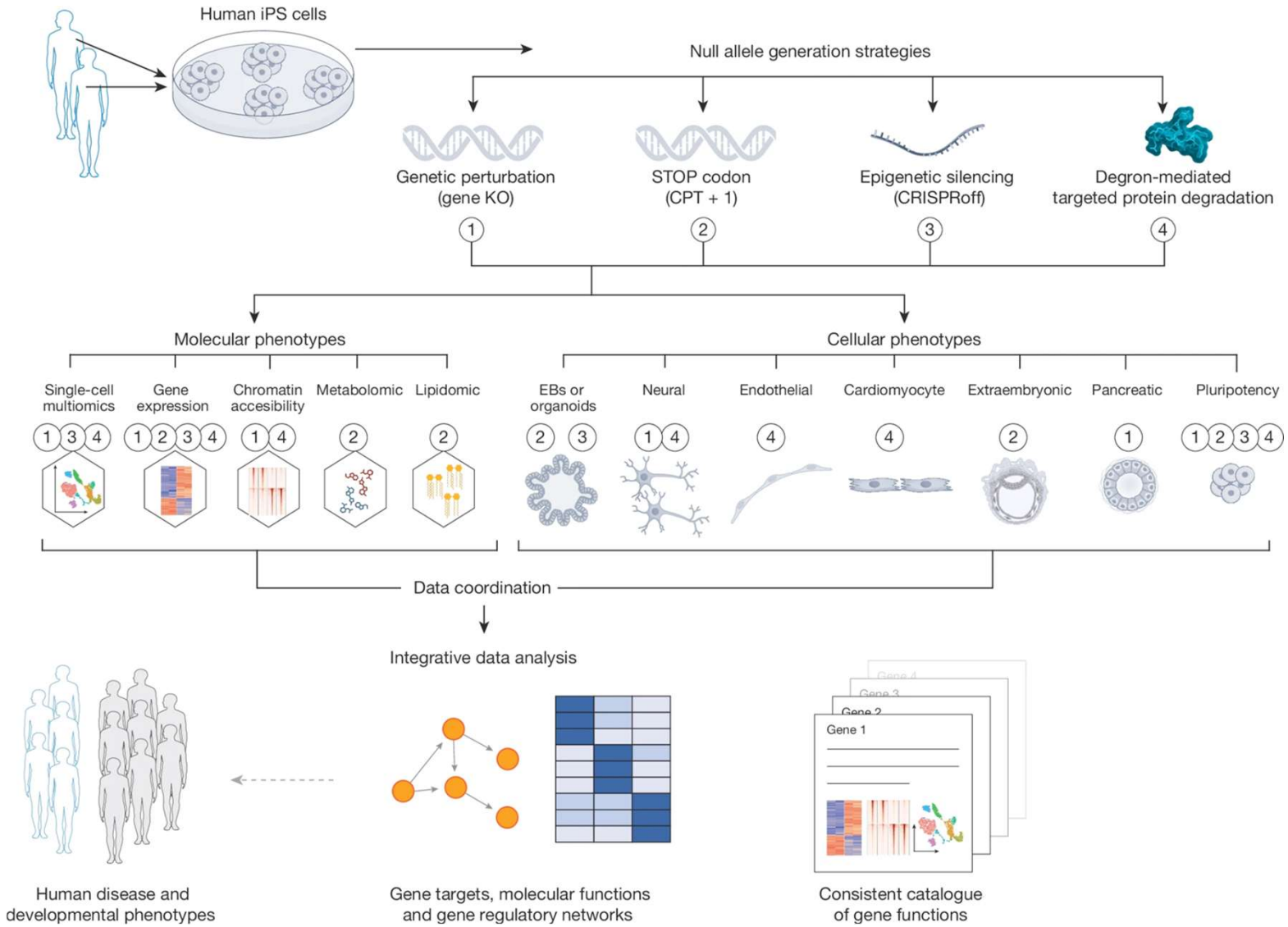
Published online: 12 February 2025

Check for updates

Mazhar Adli[1,26]✉, Laralynne Przybyla[2,26], Tony Burdett[3], Paul W. Burridge[4], Pilar Cacheiro[5], Howard Y. Chang[6], Jesse M. Engreitz[7,8], Luke A. Gilbert[9], William J. Greenleaf[7], Li Hsu[10], Danwei Huangfu[11], Ling-Hong Hung[12], Anshul Kundaje[13], Sheng Li[14], Helen Parkinson[15], Xiaojie Qiu[8,13], Paul Robson[14], Stephan C. Schürer[16], Ali Shojaie[17], William C. Skarnes[14], Damian Smedley[5], Lorenz Studer[11], Wei Sun[10], Dušica Vidović[16], Thomas Vierbuchen[11], Brian S. White[14], Ka Yee Yeung[12], Feng Yue[18], Ting Zhou[11] & The MorPhiC Consortium*

Recent advances in functional genomics and human cellular models have substantially enhanced our understanding of the structure and regulation of the human genome. However, our grasp of the molecular functions of human genes remains incomplete and biased towards specific gene classes. The Molecular Phenotypes of Null Alleles in Cells (MorPhiC) Consortium aims to address this gap by creating a comprehensive catalogue of the molecular and cellular phenotypes associated with null alleles of all human genes using in vitro multicellular systems. In this Perspective, we present the strategic vision of the MorPhiC Consortium and discuss various strategies for generating null alleles, as well as the challenges involved. We describe the cellular models and scalable phenotypic readouts that will be used in the consortium's initial phase, focusing on 1,000 protein-coding genes. The resulting molecular and cellular data will be compiled into a catalogue of null-allele phenotypes. The methodologies developed in this phase will establish best practices for extending these approaches to all human protein-coding genes. The resources generated—including engineered cell lines, plasmids, phenotypic data, genomic information and computational tools—will be made available to the broader research community to facilitate deeper insights into human gene functions.

Source: *Nature* **638**, 351–359 (2025).

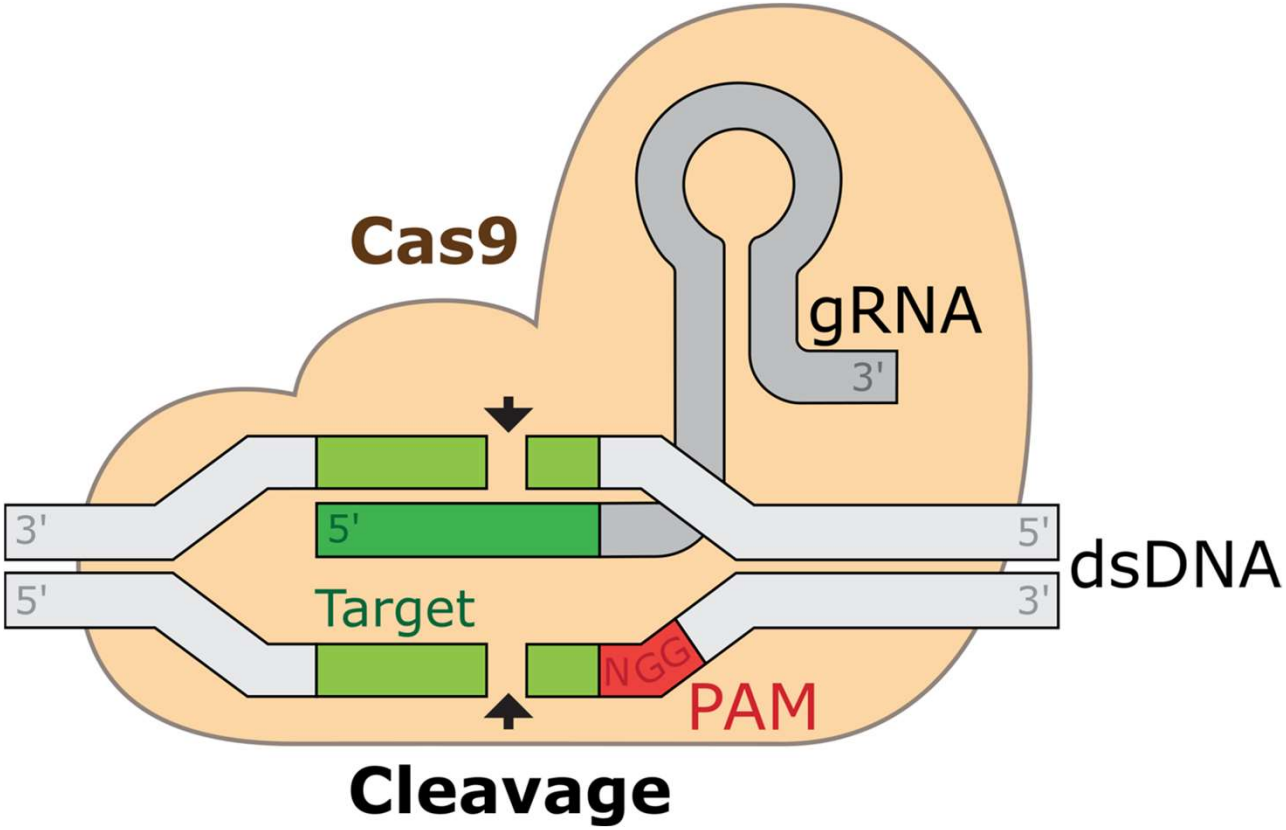


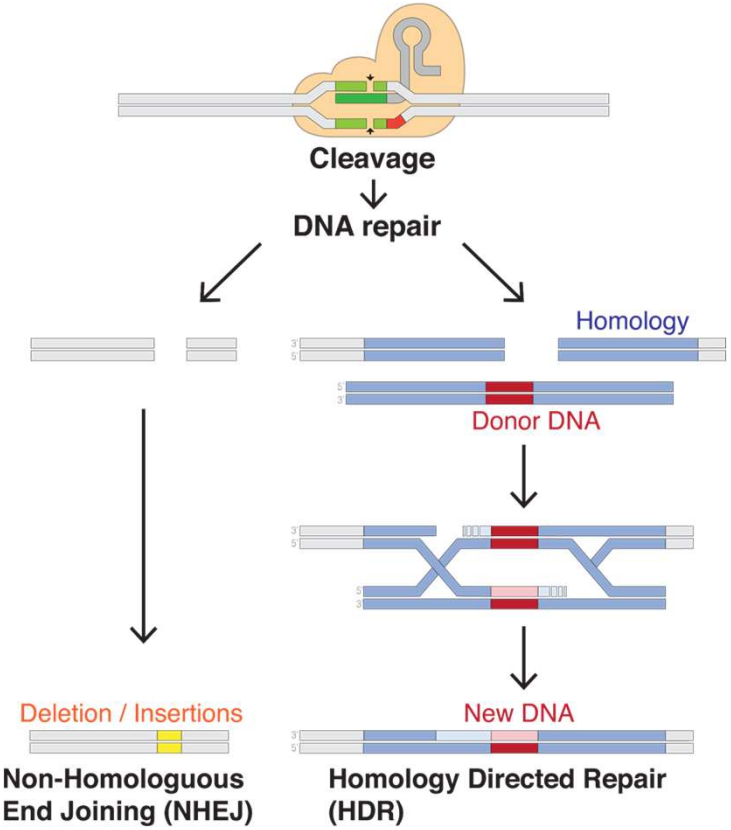
Source: *Nature* **638**, 351–359 (2025).

Fig. 2: Experimental strategies and assays. Schematic showing general experimental strategies for null-allele generation and the main cellular and molecular assays used to characterize phenotypes of null alleles.

See https://morphic.bio/methods/

Genetic engineering with Cas9





PAM: protospacer adjacent motif
gRNA: guide RNA

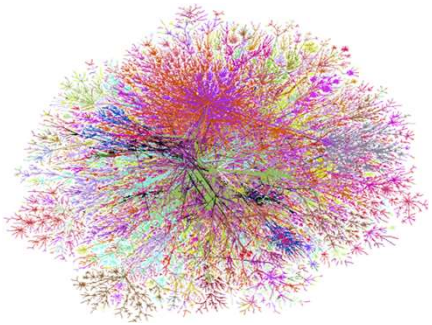
Wikipedia: CC BY-SA 4.0

Inference of gene regulatory networks (GRNs)

- DAVs will develop methods for data analysis and visualization to integratively analyse and validate MorPhiC data, thereby improving its utility for the research community.
- DAVs will use existing computational methods and/or develop new strategies to infer how genes act together in regulatory networks and pathways, e.g. approaches leveraging inference of gene-expression programs, gene regulatory networks linking transcription factors to target genes.
- DAVs may also be able to develop novel machine-learning frameworks to model how genetic perturbation affects cell states and fates via intercellular cell–cell interaction and intracellular gene regulatory networks.

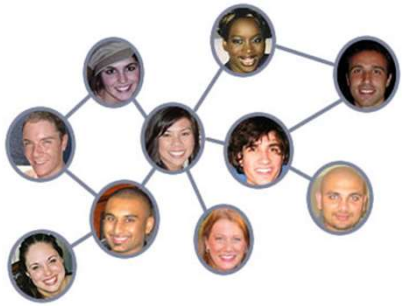
Networks as a universal language

We are caught in an inescapable network of mutuality. ... Whatever affects one directly, affects all indirectly. — Martin Luther King Jr.

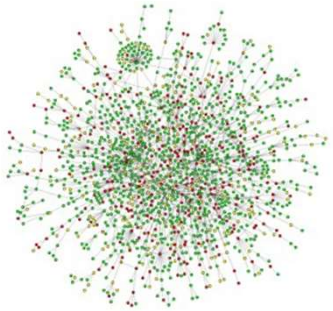


Internet

Electronic Circuit



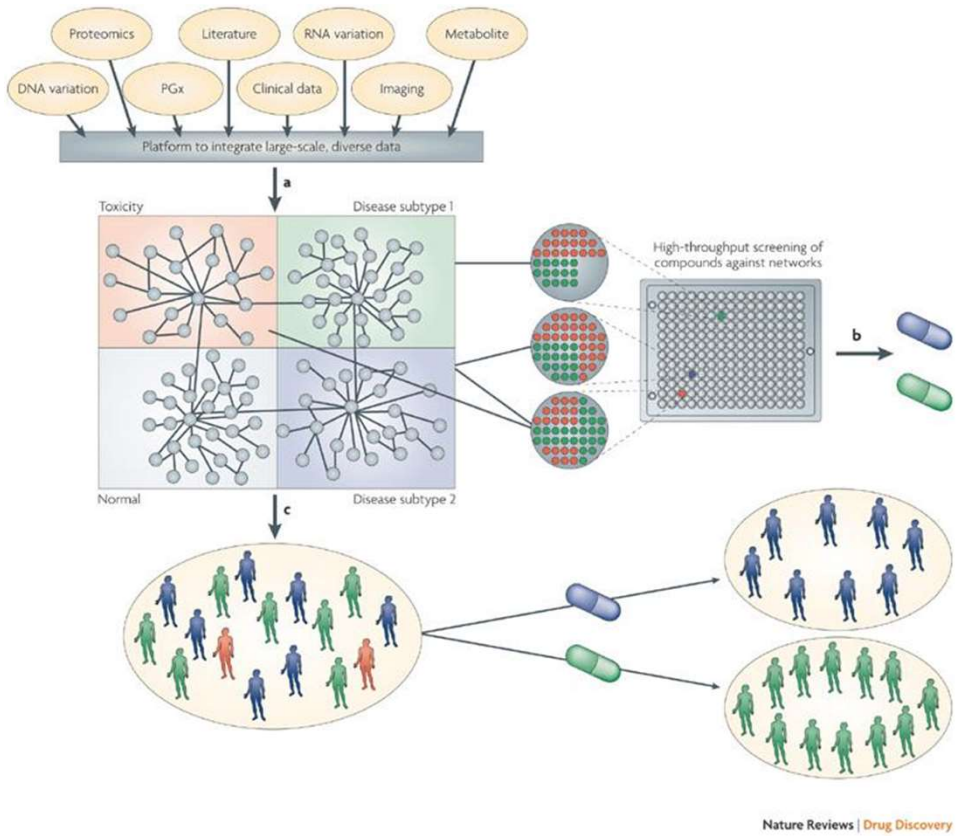
Social Network



Gene Regulatory Network (GRN)

What can we do with these molecular networks?

Network-based drug discovery



Success stories:

- Variations in DNA elucidate molecular networks that cause disease. Chen et al. Nature 2008.
- Genetics of gene expression and its effect on disease. Emilsson et al. Nature 2008.

Fig 4 Schadt et al. Nature Reviews Drug Discovery 2009

Motivation of gene network inference

- Using biochemical methods, it takes 1000's of person years to assign genes to pathways.
- Even for well-studied genomes, the majority of genes are *not* mapped to known pathways.
- As more genomes are being sequenced, and more genes are discovered, we need systematic methods to assign genes to pathways.

Wisdom of crowds for robust gene network inference

Daniel Marbach[1,2,11], James C Costello[3–5,11], Robert Küffner[6,11], Nicole M Vega[3–5], Robert J Prill[7], Diogo M Camacho[3–5,10], Kyle R Allison[3–5], The DREAM5 Consortium[8], Manolis Kellis[1,2], James J Collins[3–5,9] & Gustavo Stolovitzky[7]

Reconstructing gene regulatory networks from high-throughput data is a long-standing challenge. Through the Dialogue on Reverse Engineering Assessment and Methods (DREAM) project, we performed a comprehensive blind assessment of over 30 network inference methods on *Escherichia coli*, *Staphylococcus aureus*, *Saccharomyces cerevisiae* and *in silico* microarray data. We characterize the performance, data requirements and inherent biases of different inference approaches, and we provide guidelines for algorithm application and development. We observed that no single inference method performs optimally across all data sets. In contrast, integration of predictions from multiple inference methods shows robust and high performance across diverse data sets. We thereby constructed high-confidence networks for *E. coli* and *S. aureus*, each comprising ~1,700 transcriptional interactions at a precision of ~50%. We experimentally tested 53 previously unobserved regulatory interactions in *E. coli*, of which 23 (43%) were supported. Our results establish community-based methods as a powerful and robust tool for the inference of transcriptional gene regulatory networks.

successfully used to address many biological problems[8–11], yet when applied to the same data, they can generate disparate sets of predicted interactions[2,3].

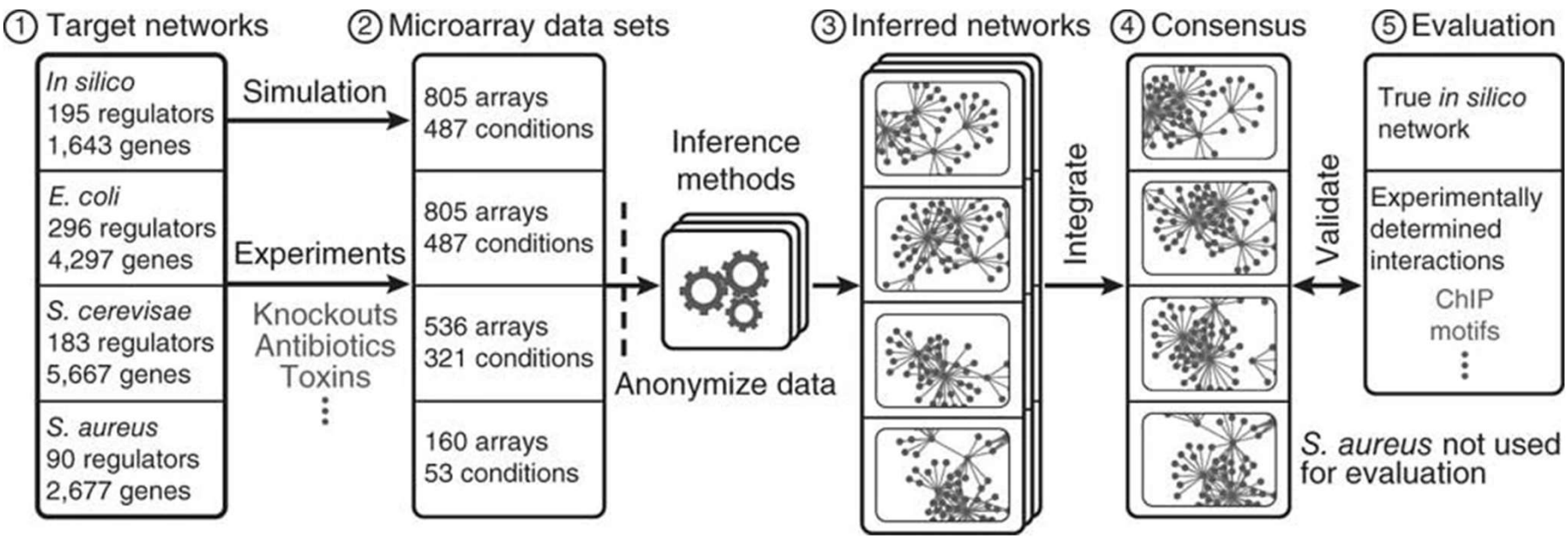
Understanding the advantages and limitations of different network inference methods is critical for their effective application in a given biological context. The DREAM project is a framework to enable such an assessment through standardized performance metrics and common benchmarks[12] (http://www.the-dream-project.org/). DREAM is organized around annual challenges, whereby the community of network inference experts is solicited to run their algorithms on benchmark data sets, participating teams submit their solutions to the challenge and the submissions are evaluated[12–14].

Here we present the results for the transcriptional network inference challenge from DREAM5, the fifth annual set of DREAM systems biology challenges. The community of network inference experts was invited to infer genome-scale transcriptional regulatory networks from gene-expression microarray data sets for a prokaryotic model organism (*E. coli*), a eukaryotic model organism (*S. cerevisiae*), a human pathogen (*S. aureus*) and an *in silico* benchmark (**Fig. 1**).

Nature methods 9.8 (2012): 796-804.

DREAM5 network inference challenge (2010)

https://www.synapse.org/Synapse:syn2787209/wiki/70350

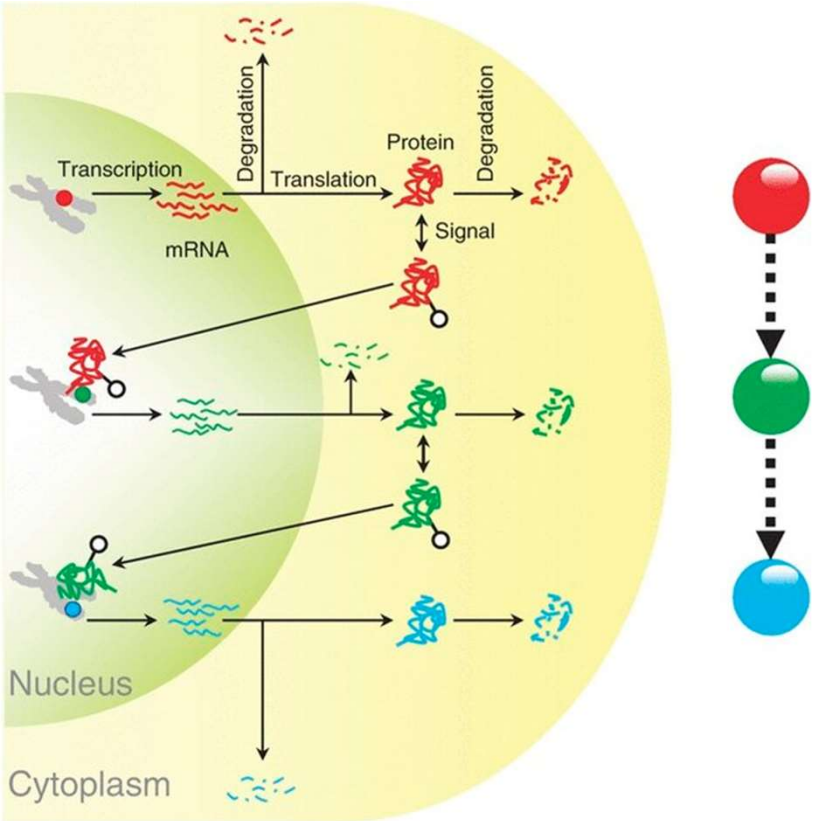


Source: **Fig 1** from *Nature methods* 9.8 (2012): 796-804.
Assessment involved the following steps (from left to right). (1) Participants were challenged to infer the genome-wide transcriptional regulatory networks of *E. coli*, *S. cerevisiae* and *S. aureus* as well as an *in silico* (simulated) network. (2) Gene-expression data sets for a wide range of experimental conditions were compiled. Anonymized data sets were released to the community with the identities of the genes hidden. (3) Twenty-nine participating teams inferred gene regulatory networks. In addition, we applied six off-the-shelf inference methods. (4) Network predictions from individual teams were integrated to form community networks. (5) Network predictions were assessed using experimentally supported interactions from *E. coli* and *S. cerevisiae* as well as the known *in silico* network.

Transcription Factors (TFs)

- TFs are "*master switches*" of the cell. They are proteins that control which genes are turned "on" or "off" by binding to specific DNA sequences.
- Humans have approximately 1,600 different transcription factors. About 8% to 10% of all human genes code for transcription factors.
- In a gene regulatory network (GRN), transcription factors are the **nodes** that control the flow of information. The DREAM5 challenge provided participants with:
  - **Expression Data:** Levels of mRNA for thousands of genes across different conditions.
  - **A List of Known TFs:** Participants were told which genes were transcription factors, but not which "target" genes they controlled.

GRN inference



- The DREAM5 challenge was to draw the **edges** (arrows) from a TF to its target gene. The goal of the challenge was to see if computational models could accurately predict which TFs regulate which target genes based solely on gene expression data.
- **Combinatorial Control:** Genes are rarely controlled by just one TF; usually several TFs is required to activate a gene.

Why are KO data useful in inferring GRNs?

- **Distinguishing Correlation from Causation**
  - In standard transcriptomics, if Gene A and Gene B both increase at the same time, they are **correlated**. However, you don't know if: Gene A activates Gene B, Gene B activates Gene A, or a third Gene C activates both of them.
  - By "knocking out" Gene A, you break the system. If Gene B stops being expressed when Gene A is gone, you have **causal proof** that A regulates B.
- **Directionality of Edges**
  - **KO data** is inherently directional. You perturb the "source" and measure the "sink." If you delete a TF and its predicted targets don't change, you know that the regulatory edge likely doesn't exist or is redundant.
- GRN inference often struggle with "false positives" (arrows that looked real but weren't). KO data can provide a "truth set" for assessment, or to help to resolve these FPs.

nature reviews genetics

https://doi.org/10.1038/s41576-023-00618-5

Review article

Check for updates

Gene regulatory network inference in the era of single-cell multi-omics

Pau Badia-i-Mompel[1], Lorna Wessels[1,2], Sophia Müller-Dott[1], Rémi Trimbour[1,3], Ricardo O. Ramirez Flores[1], Ricard Argelaguet[4] & Julio Saez-Rodriguez[1]

Abstract

The interplay between chromatin, transcription factors and genes generates complex regulatory circuits that can be represented as gene regulatory networks (GRNs). The study of GRNs is useful to understand how cellular identity is established, maintained and disrupted in disease. GRNs can be inferred from experimental data – historically, bulk omics data – and/or from the literature. The advent of single-cell multi-omics technologies has led to the development of novel computational methods that leverage genomic, transcriptomic and chromatin accessibility information to infer GRNs at an unprecedented resolution. Here, we review the key principles of inferring GRNs that encompass transcription factor–gene interactions from transcriptomics and chromatin accessibility data. We focus on the comparison and classification of methods that use single-cell multimodal data. We highlight challenges in GRN inference, in particular with respect to benchmarking, and potential further developments using additional data modalities.

Sections

Introduction

Inference of GRNs

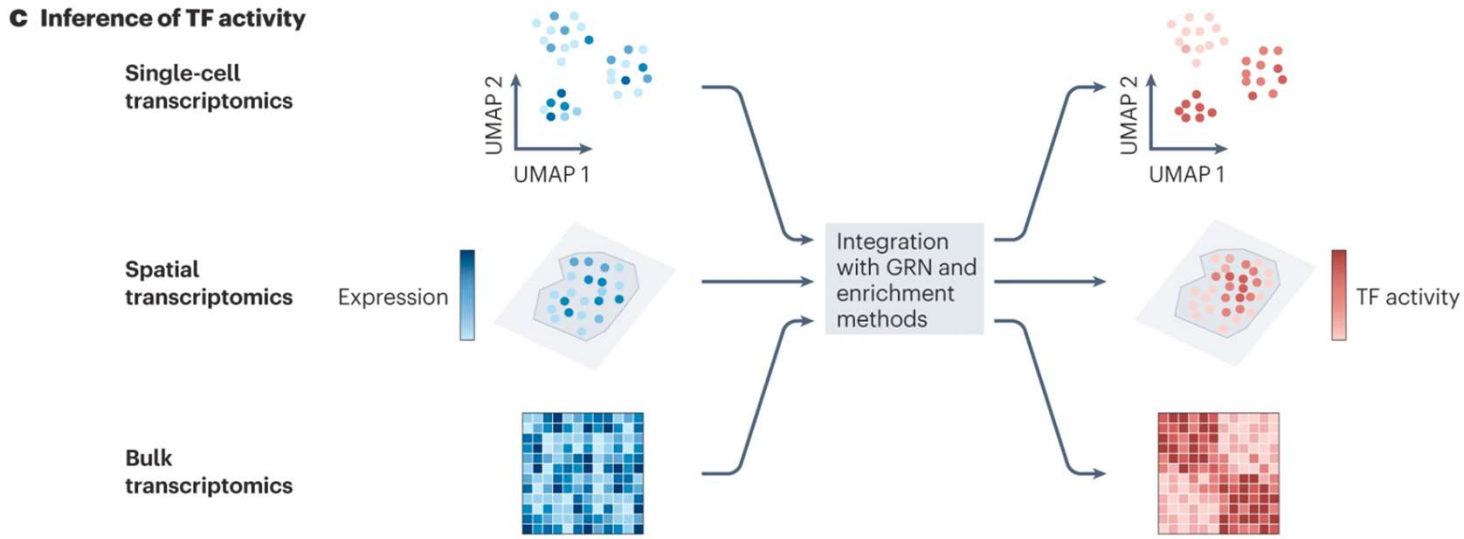
Downstream GRN analyses

Experimental assessment of GRNs

Challenges and future directions

Conclusions

Nature Reviews Genetics
24.11 (2023): 739-754.



Source: **Fig 3c**
Nature Reviews Genetics 24.11 (2023): 739-754.
GRNs can be coupled to enrichment methods to infer which TFs might be functionally active from transcriptomics data. GRNs inferred from multi-omics data can then be used to infer TF activities in other contexts, such as independent single-cell, spatial or bulk transcriptomics data.

Many many GRN inference methods in the literature

There are many benchmarking studies too.

Example: GRNBoost2

Bioinformatics, 35(12), 2019, 2159–2161
doi: 10.1093/bioinformatics/bty916
Advance Access Publication Date: 15 November 2018
Applications Note

OXFORD

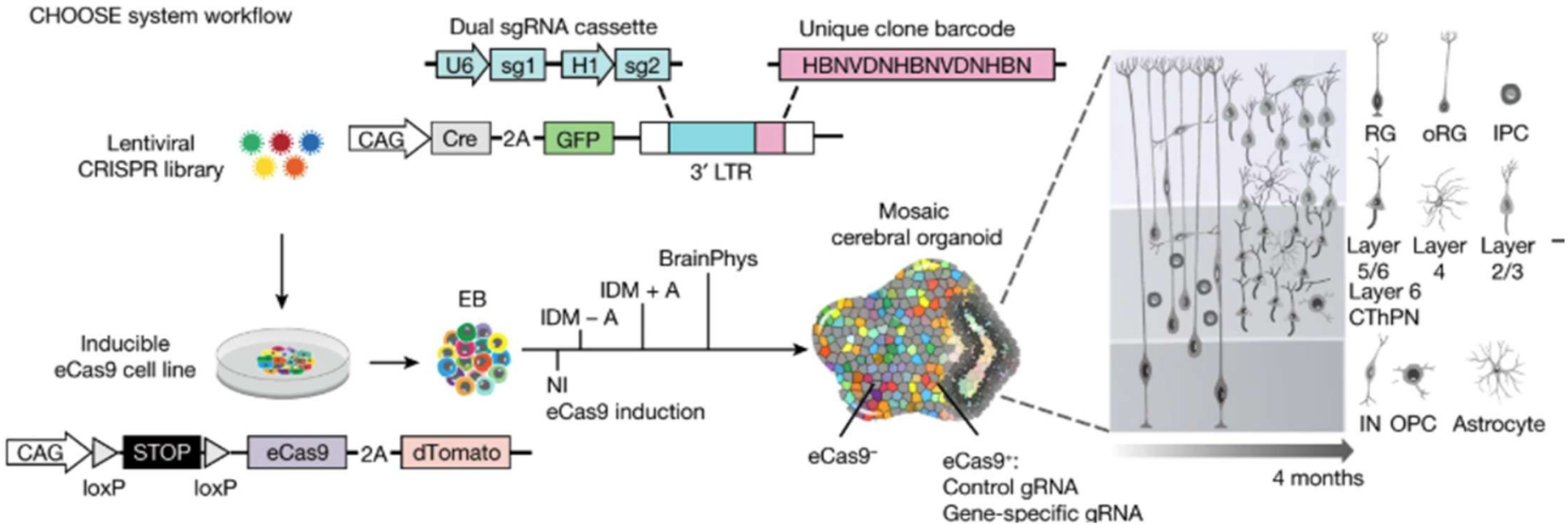
Gene expression

GRNBoost2 and Arboreto: efficient and scalable inference of gene regulatory networks

Thomas Moerman[1,2,*], Sara Aibar Santos[3,4], Carmen Bravo González-Blas[3,4], Jaak Simm[2,5], Yves Moreau[2,5], Jan Aerts[1,2] and Stein Aerts[3,4,*]

[1]KU Leuven ESAT/STADIUS, VDA-lab, [2]IMEC Smart Applications and Innovation Services, [3]Laboratory of Computational Biology, VIB Center for Brain & Disease Research, [4]Department of Human Genetics, KU Leuven, Leuven, Belgium and [5]KU Leuven ESAT/STADIUS, Bioinformatics Lab

Training data: a gene perturbation study of the developing brain

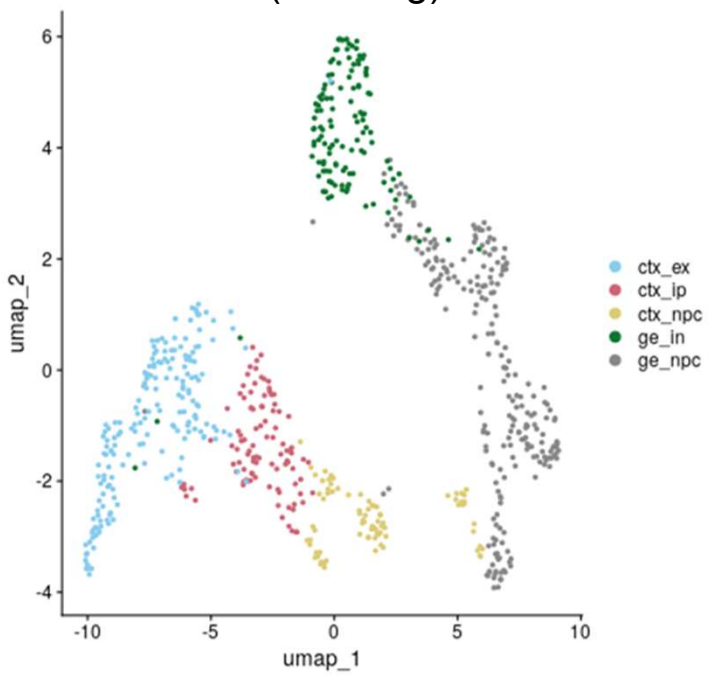


CHOOSE: CRISPR human organoids single-cell RNA sequencing

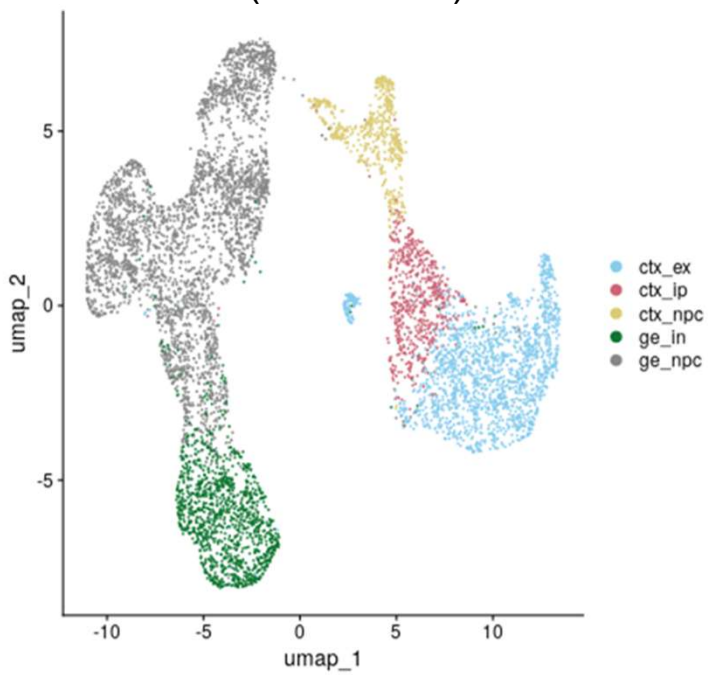
Li, *et al.* (2023) *Nature*

Overview of CHOOSE dataset

WT multiome
(Training)



WT & KO scRNA-seq
(Evaluation)



Major neuronal lineages:
1) Cortex (“ctx”) (or dorsal)
2) Ganglionic eminence (“ge”) (or ventral)

“ex” = excitatory
“in” = inhibitory

“npc” = neural progenitor cells
“ip” = intermediate progenitors

Evaluating accuracy of inferred TF targets

Hypotheses/assumptions:

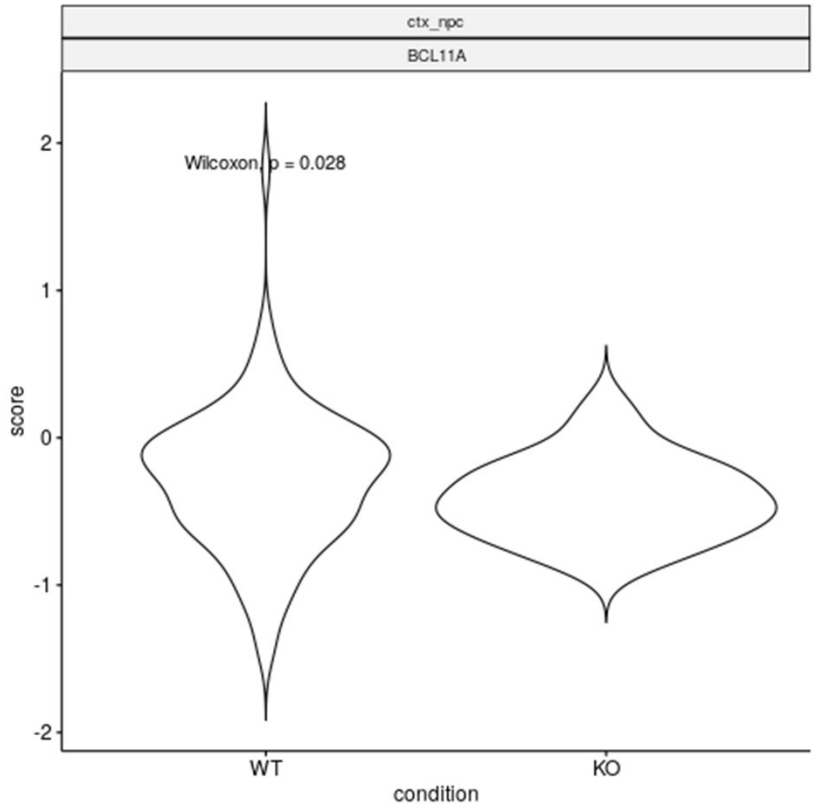
1. TF protein “activity” will be higher in WT cells than in TF KO cells.

2. We can approximate TF activity through inference with VIPER (based on expression of its putative target genes).

3. The degree of separation in computed TF activity between WT and TF KO cells is a measure of the accuracy of the TF targets used to compute the TF activity.

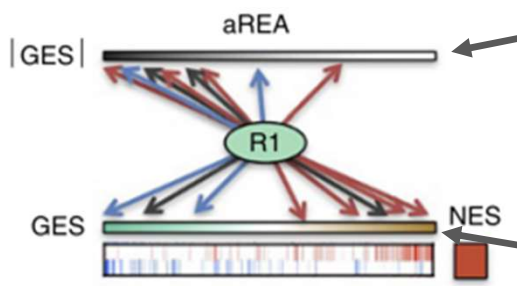
4. More accurate methods of predicting TF targets will have larger shifts in TF activity.

VIPER scores computed from GRNBoost2-predicted targets of TF BCL11A in ctx_npc cells



Predicting TF activity with VIPER

High TF activity

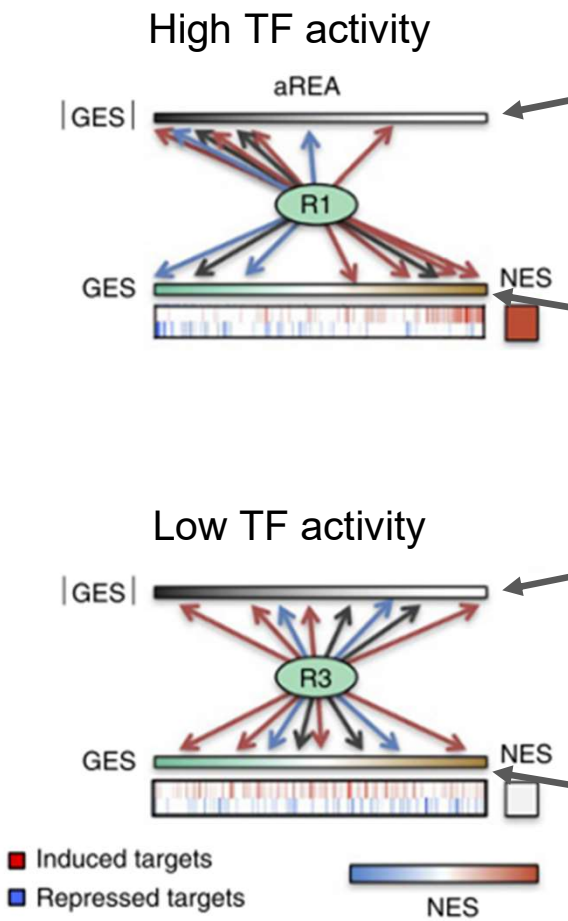


1-tail test
Predicted (induced & repressed) targets are near the expression extremes

2-tail test
Predicted induced targets (red) have high relative expression
Predicted repressed targets (blue) have low relative expression

Alvarez, *et al.* (2016) *Nat Genet*

Predicting TF activity with VIPER



1-tail test
Predicted (induced & repressed) targets are near the expression extremes

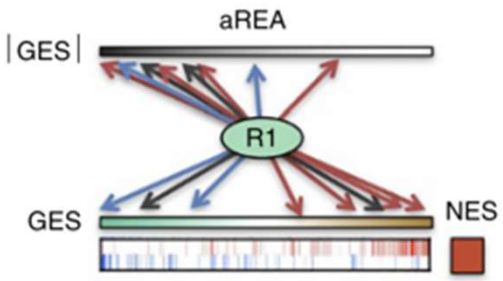
2-tail test
Predicted induced targets (red) have high relative expression
Predicted repressed targets (blue) have low relative expression

1-tail test
Predicted (induced & repressed) targets are distributed throughout expression spectrum

2-tail test
Predicted (induced & repressed) targets are distributed throughout expression spectrum

Alvarez, *et al.* (2016) *Nat Genet*

The mechanics of VIPER



Definitions

g_i: expression of gene *i*

MoR_i: Mode of Regulation of gene *i*

from strong repressed (-1) to strongly induced (1)

2-tail test

Test for high expression of induced targets & low expression of repressed targets

$$q_{2,i} = \frac{\mathrm{rank}\,(\mathrm{g_i})}{n+1} \in \left[\frac{1}{n+1}, \frac{n}{n+1}\right]$$

$$\Phi^{-1}\left(q_{2,i}\right) \in \left(-\infty, \infty\right)$$

$$S_2 \equiv \sum_i \mathrm{MoR}_i\ \Phi^{-1}\left(q_{2,i}\right)$$

1-tail test

Test for extremal expression of all targets

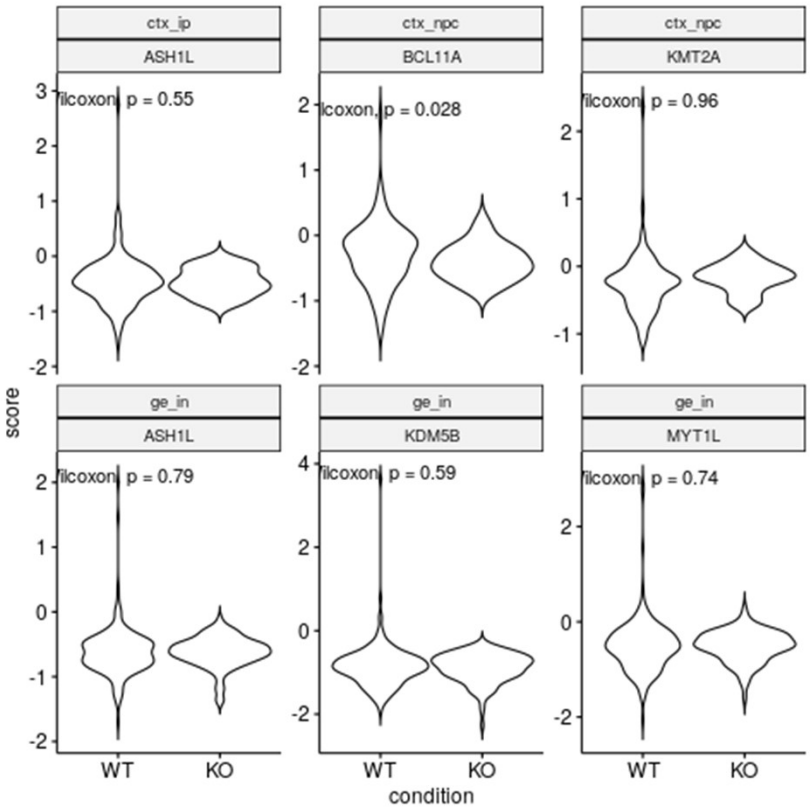
$$q_{1,i} = 2\left|q_{2,i} - 0.5\right| + \mathrm{const}$$

$$S_1 \equiv \sum_i \left[1 - \left|\mathrm{MoR}_i\right|\right]\ \Phi^{-1}\left(q_{1,i}\right)$$

3-tail test

$$S_3 = \begin{cases} \mathrm{sign}\,(S_2)\left[|S_2| + S_1\right] & \text{if } S_1 > 0 \\ \mathrm{sign}\,(S_2)\,|S_2| = S_2 & \text{if } S_1 < 0 \end{cases}$$

A broader testbed for evaluation



We anticipate there is adequate data (e.g., # of cells) to predict TF targets in six TF / cell type combinations (shown on the left).

However, of these, GRNBoost2 was only able to predict meaningful targets for TF BCL11A in the ctx_npc cell type.

Goals:

1. Develop an approach that achieves a larger separation of VIPER scores for BCL11A / ctx_npc (i.e., smaller p-value).
2. Develop an approach that achieves a significant separation of VIPER scores for any of the other combinations.